



Multi-step framework for glaucoma diagnosis in retinal fundus images using deep learning

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Abstract

Glaucoma is one of the most common causes of blindness in the world. Screening glaucoma from retinal fundus images based on deep learning is a common method at present. In the diagnosis of glaucoma based on deep learning, the blood vessels within the optic disc interfere with the diagnosis, and there is also some pathological information outside the optic disc in fundus images. Therefore, integrating the original fundus image with the vessel-removed optic disc image can improve diagnostic efficiency. In this paper, we propose a novel multi-step framework named MSGC-CNN that can better diagnose glaucoma. In the framework, (1) we combine glaucoma pathological knowledge with deep learning model, fuse the features of original fundus image and optic disc region in which the interference of blood vessel is specifically removed by U-Net, and make glaucoma diagnosis based on the fused features. (2) Aiming at the characteristics of glaucoma fundus images, such as small amount of data, high resolution, and rich feature information, we design a new feature extraction network RA-ResNet and combined it with transfer learning. In order to verify our method, we conduct binary classification experiments on three public datasets, Drishti-GS, RIM-ONE-R3, and ACRIMA, with accuracy of 92.01%, 93.75%, and 97.87%. The results demonstrate a significant improvement over earlier results.

Keywords Fundus image · Glaucoma diagnosis · Multi-step · Vessel elimination · RA-ResNet

1 Introduction

Glaucoma is the second leading cause of blindness worldwide, causing increased intraocular pressure and irreversible damage to nerve fibers, resulting in optic disc (OD) and optic cup (OC) lesions. Because vision loss due to glaucoma is irreversible, early screening and treatment of this chronic disease is essential to prevent irreversible vision loss, and blindness can be prevented with treatment in the initial stage [1].

The diagnosis of glaucoma depends on information from various clinical examinations, including visual field (VF), optical coherence tomography (OCT), and fundus images. OCT and VF are expensive, and their public dataset are not yet widely available [2], whereas fundus images are easy to collect and are widely used for the diagnosis of fundus diseases; therefore, screening glaucoma by fundus retinal images has become the most common method at present [3]. Nowadays, the use of computer aided diagnosis (CAD) technology for glaucoma diagnosis has become a hot spot for research, in particular, deep learning technology has attracted wide attention due to its high accuracy.

With the rapid development of deep learning, its application in the aided diagnosis of glaucoma has made great progress. In the early stage, the research of glaucoma diagnosis based on fundus images was mainly realized by directly inputting glaucoma data into common networks, for example, Rao [4], Serener [5], and Chen et al. [6] employ this method for glaucoma diagnosis. Although these methods have simple process, there are several problems:

Because their process of simply feeding data into the network does not emulate the ophthalmologist's diagnostic

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process, which combines pathological knowledge, such as focusing on the optic disc region and considering vascular interference, they lack the ability to extract enough pathological features. Therefore, more and more researchers have constructed multi-step models based on the pathological knowledge of glaucoma. For example, aiming at the importance of the optic disc region, Bajwa proposed a model which firstly locates and extracts such region of fundus images, and then used deep learning networks to classify the optic disc region images [7]. Allan also used a similar method with good diagnostic results [8]. Later, Chai believed that in addition to the optic disc region, other regions of the fundus image also had lesion information, so they designed a model named MB-NN, which fused the fundus image, optic disc region image, and other pathological features and significantly improved the classification performance of glaucoma [9]. For ophthalmologists, such areas are called regions of interest (ROIs), which contain rich pathologic features that require more attention. And since these models combine pathological knowledge, they can better extract pathology features for diagnostic results than the method directly inputting fundus images into the networks, indicating that using enough pathological knowledge to build a model with stronger feature extraction ability will improve the diagnostic performance of glaucoma. The above methods, despite combining partial pathological knowledge, ignored vascular interference. Meier concluded that blood vessels in retinal images interfere with the diagnosis of glaucoma [10]. If this factor is added to the glaucoma diagnostic model, the diagnostic effect of glaucoma can be greatly improved. As far as we know, there are no deep learning models that take this into account.

Glaucoma fundus images have high resolution and rich feature information, requiring the networks to have strong feature extraction capability. Besides, a common problem in the current medical imaging field, especially in the field of glaucoma diagnosis, is that it is difficult to obtain a large amount of data. However, the above researchers did not design a network with stronger feature extraction capability based on the characteristic of glaucoma data. Actually, designing more targeted networks for image characteristics and data volume problems is a commonly used method for medical images at present. For example, Li aimed at the characteristics of chest X-ray images and proposed a network called Cov-Net [11], which enhanced the extraction of chest X-ray image nuances and key details. Yi proposed an MS module [12] as a new functional module for the characteristics of rectal cancer images based on ultrasound images, which can extract multi-scale information from rectal cancer images more efficiently. In addition to this, Wang [13], He [14], and Xu et al. [15] also designed deep learning networks based on the characteristics of different medical images in order to better accomplish the diagnostic task. Therefore,

designing the proper network focusing on the characteristics of each image can greatly improve the diagnostic efficiency.

Based on the above discussion, we can draw the following conclusions: (1) Due to the low cost, simple operation, high image resolution, and large amount of information, the automatic glaucoma diagnosis model based on fundus images is more practical. (2) The combination of deep learning and glaucoma pathological knowledge to build a glaucoma diagnostic model can achieve better results. Some models have considered the optic disc information and fundus information. (3) For the small amount of fundus datasets and high resolution and rich feature information of fundus images, in order to effectively improve the diagnostic performance of glaucoma, a new deep learning network with stronger feature extraction ability is required. On this basis, if vessel interference is taken into account, the diagnostic effect will be better improved.

In this paper, we propose a new multi-step framework named MSGC-CNN (Multi-Step Glaucoma Classification based on convolutional neural network) for glaucoma diagnosis to make full use of the knowledge of glaucoma pathology and extract sufficient features from images. The main contributions of the framework are as follows: (1) In view of the characteristics of fundus images, such as small amount of data, high resolution, and rich feature information, we design a new feature extraction network RA-ResNet, which can not only extract sufficient feature information from fundus images, but also avoid over-fitting due to the complexity of the network. (2) Aiming at the problem that the blood vessels in the optic disc can interfere with the diagnosis of glaucoma, we add a vessel elimination step in the framework and use U-Net to perform vascular segmentation, to eliminate the interference of the information in the optic disc region. (3) We fuse the features of the original fundus image and optic disc image without vascular interference to obtain richer feature information and finally perform glaucoma classification based on the fused features.

The rest of the paper is arranged as follows: Section 2 describes the process of our proposed MSGC-CNN diagnostic model in detail. Section 3 describes our experimental parameter settings, network comparison experiments, ablation experiments performed on multiple datasets, and compares the results with state-of-the-art methods. Finally, the work of this paper is summarized in Section 4.

2 Method

According to the characteristics of glaucoma, the proposed multi-step framework MSGC-CNN integrates optic disc region information, vascular interference, and fundus information which leads to better glaucoma diagnosis. The framework is shown in Fig. 1 which consists of four steps: the first

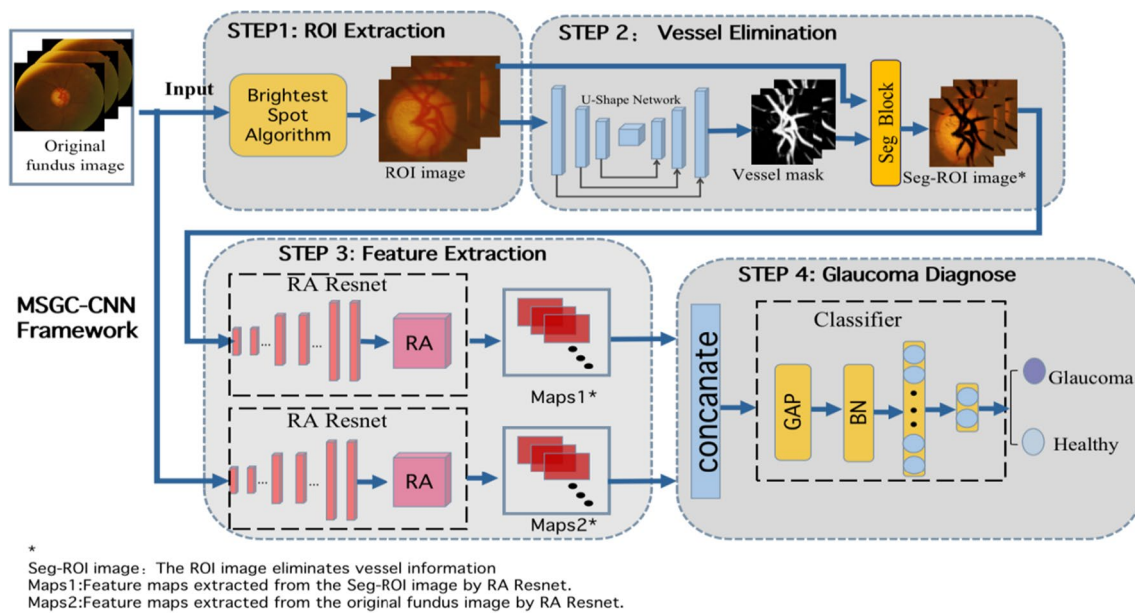


Fig. 1 Framework of the proposed MSGC-CNN for glaucoma classification with multiple steps

step is ROI extraction, using a ROI extraction algorithm to locate and extract ROI; the detailed definition of the ROI is given in Section 2.1. The second step is vessel elimination, using a segmentation algorithm to extract the vessel mask, so that the vessel information in the optic disc region can be located and eliminated. The third step is feature extraction, where a new designed network RA-ResNet is used to extract the features of fundus image and optic disc region image. The fourth step is glaucoma diagnosis, which is completed by fusing the two feature maps into the classifier.

2.1 Step1: ROI extraction

Glaucoma disease information is mainly concentrated in the optic disc region, based on which neural networks can extract more effective feature information. Therefore, we define the small region containing the optic disc as the ROI in this paper. As shown in step 1 in Fig. 1, we adopt a Brightest Spot Algorithm [16] to extract the ROI area of the original fundus image. Firstly, the algorithm preprocesses the

image with Gaussian blur, then locates the brightest point in the optic disc, and finally extracts the ROI image with appropriate size based on the brightest point. Figure 2 shows the original fundus images and the corresponding ROI images from the Drishti-GS dataset and RIM-ONE-R3 dataset.

2.2 Step 2: Vessel elimination

The disease information of glaucoma is mainly concentrated in the ROI region mentioned in step 1, but the image obtained by step 1 still contains vessel information, which will interfere with the diagnosis of glaucoma. Therefore, in step 2, we perform vascular elimination on ROI images. In this paper, we use U-Net [17] as a vascular segmentation network. The details of the network structure are shown in Fig. 3. The network consists of an encoder and a decoder. The activation function after each convolutional layer is ReLU, and a dropout of 0.2 is used between two consecutive convolutional layers. We also used UnSampling2D for unsampling.

Fig. 2 Examples of the original fundus images and the corresponding ROI images: **a** an original fundus image from Drishti-GS dataset. **b** The corresponding ROI image of fundus image 1. **c** An original fundus image from RIM-ONE-R3 dataset. **d** The corresponding ROI image of fundus image 2

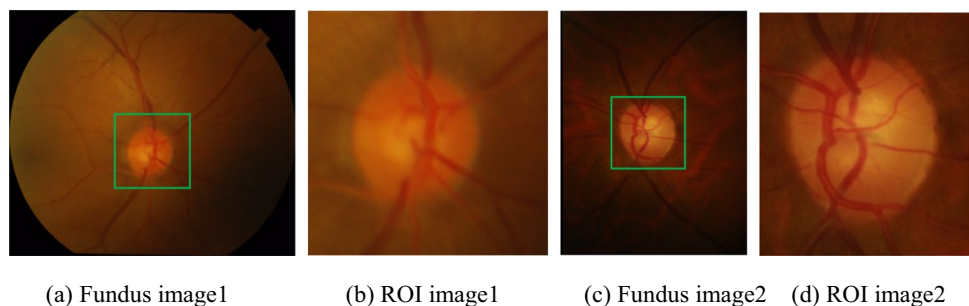
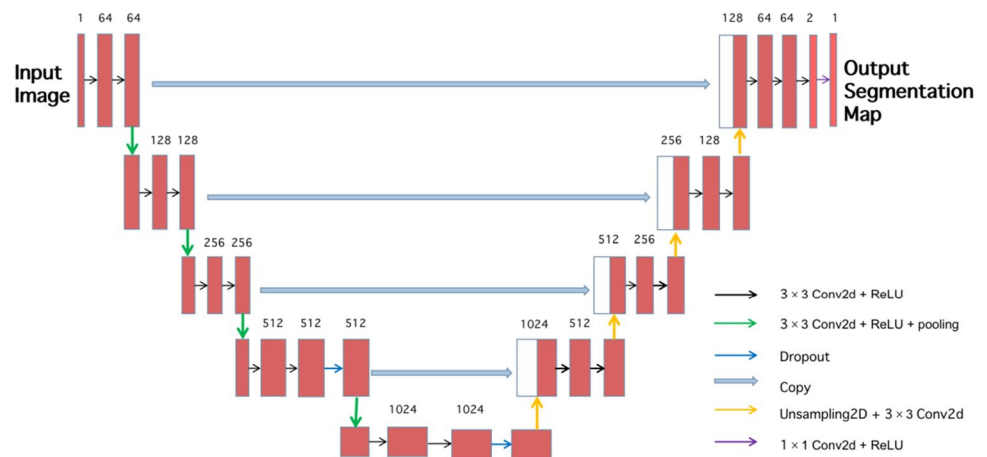


Fig. 3 U-Net architecture

As shown in step 2 of Fig. 1, in order to obtain the ROI image without vascular interference, we first preprocess the ROI image with green channel extraction, histogram equalization, standardization, and gamma transformation to highlight the vessel contour information, then input the preprocessed image into U-Net. Thus, the vessel mask of the ROI image is obtained. Subsequently, we input the ROI image and vessel mask into the Seg Block to get the Seg-ROI image. The details of the operation of Seg Block are shown in Eq. (1):

$$R_{\text{seg}} = R - R \cdot M_{\text{ves}} \quad (1)$$

where R_{seg} presents the ROI image without vascular interference (i.e., Seg-ROI). R presents the preprocessed ROI image, and M_{ves} is the vessel mask of the ROI image output by U-Net. Figure 4 shows examples of ROI image, vessel mask, and Seg-ROI image.

2.3 Step 3: Feature extraction

As shown in step 3 of Fig. 1, we extract features from both the original fundus image and Seg-ROI image. In this step, because the glaucoma dataset is very small, the network

is required to have a very strong feature extraction ability to extract enough pathological information. RA-EfficientNet in literature [18] adds the feature enhancing module RA (residual attention) module to the EfficientNet so that it can better extract the features in fundus images, thus achieving good results on the APTOS 2019 dataset. However, the effect of the network on glaucoma datasets based on fundus images is not ideal, because the too deep and too wide EfficientNet is easy to overfit on small datasets. To address this issue, we enhance the ResNet50 architecture by integrating an RA module, creating a new network dubbed RA-ResNet. We then employ transfer learning techniques with this modified network as the backbone for feature extraction. The specific implementation is as follows: After the original image is preprocessed by normalization and histogram equalization, it is input into ResNet50* pre-trained on ImageNet, and then the obtained results are input into RA block to obtain enhanced feature maps as output. As shown in step 3 of Fig. 1, Maps1 are the features extracted from the original image, and Maps2 are the features extracted from the Seg-ROI image. The overall structure is shown in Fig. 5, and each functional block will be elaborated in detail as follows.

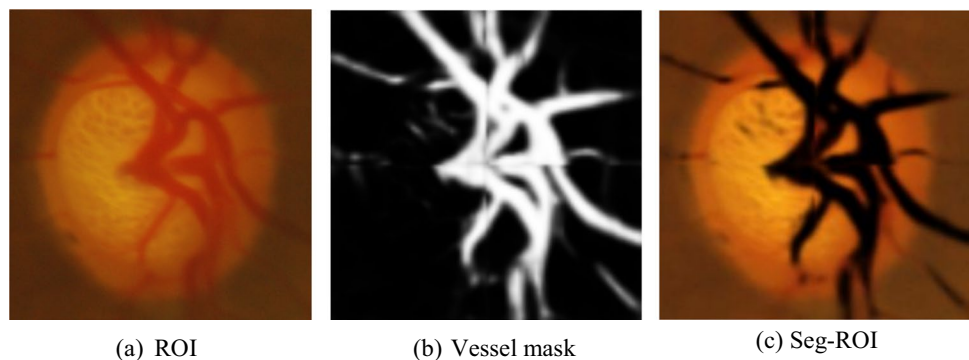
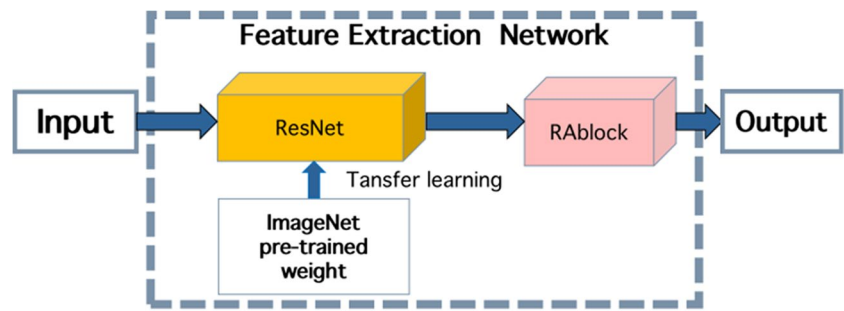
Fig. 4 Examples of three types of image used in Step 2: **a** ROI image. **b** Vessel mask. **c** Seg-ROI image

Fig. 5 The structure of feature extraction network

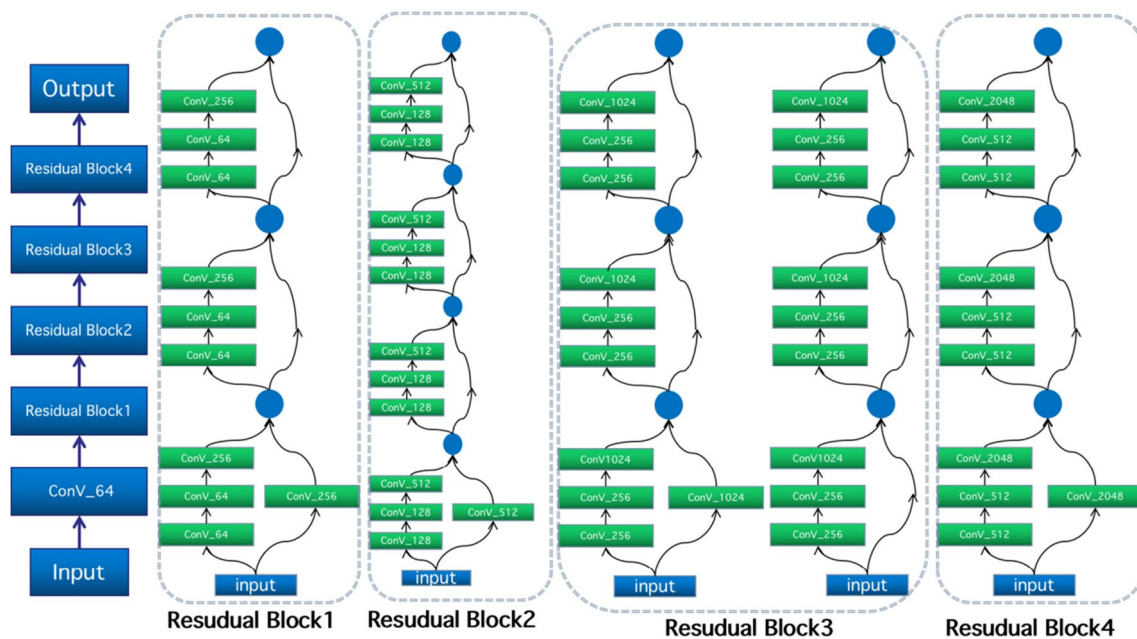
* Pre-trained weight of ResNet50 are provided by Tensorflow.

2.3.1 ResNet50

Glaucoma fundus image data has the characteristic of having a small amount of data but rich feature information. For such images, overly deep networks are prone to overfitting, while shallow networks may not fully extract feature information. “ResNet50” falls between the two and can achieve excellent results on small datasets thus becoming our ultimate choice. As shown in Fig. 6, ResNet50 [19] contains four residual convolution blocks: Residual Block1, Residual Block2, Residual Block3, and Residual Block4. The design of residual blocks reduces the problem of vanishing gradients in the network, so as to better obtain high-level features.

2.3.2 RA block

Figure 7 shows the structure of RA block [18], which is specifically designed for fundus image features. Fundus images contain a large amount of useful feature information due to their high resolution. To extract this information more sufficiently, and inspired by ResNet [19] and GCNet [20], the module combines the depthwise separable convolution, residual mechanism, and GC attention mechanism, among which GC attention can better extract global information and residual mechanism can extract more detailed information. Therefore, in the task of glaucoma classification based on fundus images, we adopt RA block to achieve more excellent results.

**Fig. 6** ResNet50 architecture

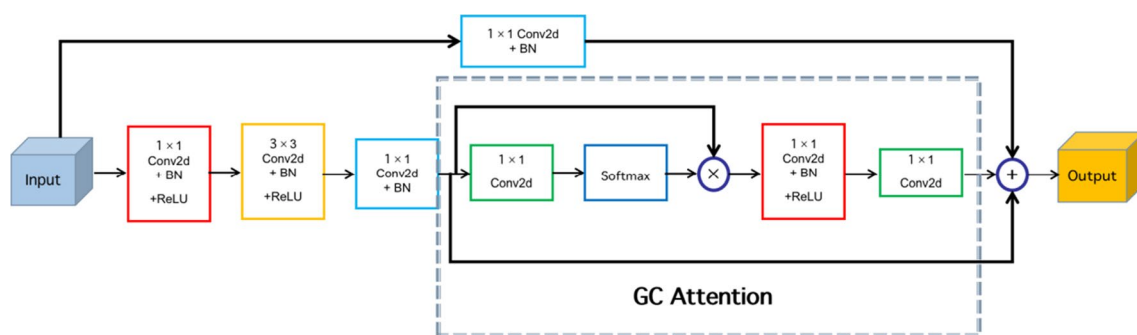


Fig. 7 The structure of the RA block

2.3.3 Transfer learning

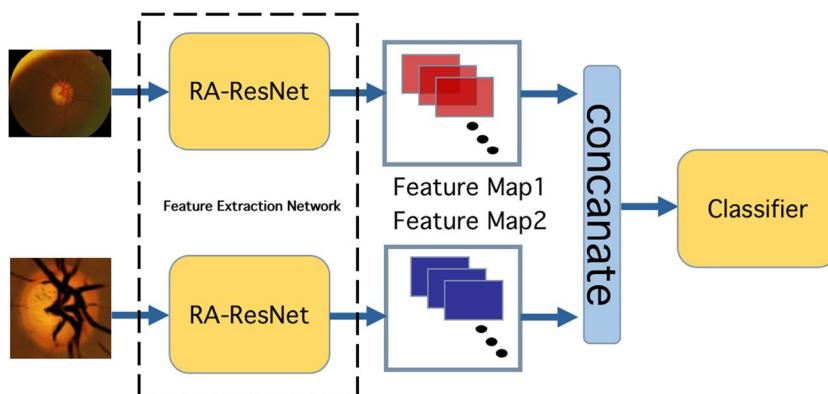
Since there are not enough public datasets, it is difficult to obtain satisfactory results based on deep learning techniques. To solve this problem, transfer learning technology is adopted in our model, and the steps are as follows: (1) pre-training network. In order to obtain the pre-trained network, the network is first pre-trained on ImageNet, which is currently the largest image recognition dataset in the world, including 1.2 million images with 1000 categories. The pre-training on large datasets can obtain rich features. (2) The glaucoma data are input into the pre-trained ResNet50 to obtain the feature maps.

2.4 Step 4: Glaucoma diagnose

The pathological features of glaucoma lesions were mainly concentrated in the optic disc area, but there were also some useful features outside the optic disc, so we performed feature fusion in this step. As shown in Fig. 8, after we extract Maps1 and Maps2 in step 3, we fuse the feature maps of the two branches. The fusion method is shown in Eq. (2):

$$F_{\text{fusion}} = \text{Concanate}(F_1, F_2) \quad (2)$$

Fig. 8 The structure of feature extraction and fusion



where F_{fusion} represents the fused feature maps, F_1 represents the feature maps extracted from the Seg-ROI image namely Maps1, and F_2 represents the feature maps extracted from the original fundus image namely Maps2.

Finally, we input F_{fusion} into classifier to obtain the diagnosis of glaucoma. The structure of the classifier is shown in Step4 in Fig. 1. The new classifier consists of a Global Average Pooling (GAP) layer, a BN layer, a fully connected layer, and a Softmax layer. The GAP layer reduces the spatial resolution, the BN layer can enhance the convergence of the network, and the last Softmax layer is used for fundus image classification task.

3 Experiments results and discussion

In this section, in order to verify the effectiveness and superiority of the proposed MSGC-CNN framework, we set up three sets of experiments: (1) In order to verify the performance of RA-ResNet proposed in this paper, we set comparison of different CNN networks and RA Block. (2) In order to verify the effectiveness of each step of the proposed algorithm, we conducted ablation experiments. (3) In order to verify the superiority of the proposed algorithm, we compared the proposed algorithm with methods in other advanced literatures.

3.1 Dataset

We used four different public datasets to validate the proposed method, among which the DRIVE dataset was used to train the vascular segmentation model, and the ACRIMA dataset, RIM-ONE-R3 dataset, and Drishti-GS dataset were used to classify glaucoma.

DRIVE dataset [21] The DRIVE dataset was derived from the Diabetic Retinal Damage Detection Project in the Netherlands. The dataset has 40 retinal images with a size of 584×565 . There are 20 images in the training set and 20 images in the test set. Each color fundus image has a gold standard for vascular segmentation.

ACRIMA dataset [22] The ACRIMA dataset was founded by the Ministerio de Economía y Competitividad of Spain. The database contained 705 fundus images, including 396 for glaucoma patients and 309 for normal.

RIM-ONE-R3 dataset [23] The RIM-ONE-R3 dataset contains 159 retinal images (85 healthy and 74 glaucoma). All images were captured using a Kowa WX 3D non-STEREO camera with a resolution of 2144 by 1424 pixels. The diagnosis of each image in the dataset was performed by two experts.

Drishti-GS dataset [24] Drishti-GS is a retinal image dataset evaluated by four experts. It provides a set of retinal images for both normal and glaucoma eyes. It consists of 101 color fundus images, including 31 for normal eyes and 70 for glaucoma.

3.2 Implementation details

All experiments were performed on computers equipped with Intel (R) Core (tm) i9-10900 k cpu@3.7 GHz, 64.0 GB of ram, NVIDIA GEFORCE GTX 3080 and 64-bit Windows10 operating systems. The compiler is Pycharm3.6.0, and all experiments are based on TensorFlow2.4.0 framework. In this paper, all the classification experiments were performed using fivefold cross validation. In each fold, 80% of the data is trained, and the remaining 20% of the data is tested to get the results. We combined the results of each fold to get the mean value as the final results. The parameters of our model are set as follows: the size of the image input to the network is 224×224 , the network optimizer is set as ADAM, the learning rate is set as 0.001, the loss function is set as cross-entropy, batch size is set as 16, and the epoch is set as 50.

3.3 Evaluation metrics

The evaluation metrics considered in our approach to assess the classifier performance are accuracy, precision, kappa, F1-score, and area under the curve (AUC):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Kappa} = \frac{\text{Observed Accuracy} - \text{Expected Accuracy}}{1 - \text{Expected Accuracy}} \quad (5)$$

$$F1 - \text{Score} = \frac{2 * TP}{(2 * TP) + FP + FN} \quad (6)$$

3.4 Experiment results

In this part, experiments are designed to verify the efficiency and superiority of the proposed MSGC-CNN framework on different datasets.

3.4.1 Comparison of different CNN networks and RA block

To verify the performance of RA-ResNet proposed in this paper, in our initial experiments based on fundus image, we indeed tested and validated multiple networks. In the end, considering many factors, including previous research findings, performance metrics, and alignment with our research objectives, we finally selected ResNet50 [19], Xception [25], EfficientNet [26], and MobileNetV2 [27] as the comparison models because they have been widely used and demonstrated good performance in fundus images. We compared the proposed algorithm with several widely used networks in fundus images, such as ResNet50 [19], MobileNetV2 [25], Xception [26], and EfficientNet [27] in three datasets, namely, Drishti-GS, RIM-ONE-R3, and ACRIMA. At the same time, in order to verify the performance of RA block, we compared the diagnostic effects of each network before and after adding RA block.

Tables 1, 2, and 3 respectively show the performance of each network on the Drishti-GS dataset, RIM-ONE-R3 dataset, and ACRIMA dataset. Table 1 shows the classification effect of each deep learning network in the Drishti-GS dataset. It can be seen from the results: (1) RA-ResNet achieves the best classification effect among all networks. The accuracy reached 86.57%. The precision reached 87.45%. The F1-Score reached 87.06%. The kappa reached 73.33%. In terms of AUC, it reached 87.36%. (2) Among Xception, EfficientNet, MobileNetV2, and ResNet50 networks, ResNet50 network is better than other networks in accuracy, precision, F1-score, kappa, and AUC. This indicates that ResNet50 is the most suitable for glaucoma diagnostic models based on fundus images among these

Table 1 Comparison of different CNN networks and RA block in Drishti-GS dataset

Model	Accuracy (%)	Precision (%)	F1-score (%)	Kappa (%)	AUC (%)
Xception	73.52	75.98	74.19	43.54	73.32
EfficientNet	78.57	78.57	78.57	78.57	57.14
MobileNetV2	82.14	82.39	82.17	64.29	82.31
ResNet50	85.71	85.71	85.71	71.43	81.74
Xception + RA	76.19	77.55	76.69	44.44	73.33
EfficientNet + RA	79.41	84.98	70.50	55.09	82.44
MobileNetV2 + RA	82.35	85.96	83.02	61.94	84.58
ResNet50 + RA	86.57	87.45	87.06	73.33	87.36

For the best result in the table of each evaluation metric, the font is set to bold. The same goes for the table that follows

deep learning networks. For glaucoma fundus images, overly deep networks like EfficientNet are prone to overfitting, while shallow networks like MobileNet and Xception may not fully extract feature information. ResNet50 falls between the two, so it works best. (3) After the addition of RA block, the performance of each network has been improved to some extent, because the RA block improves the attention to the lesion area.

Tables 2 and 3 show the performance of RIM-ONE-R3 and ACRIMA datasets on various networks, among which the RA-ResNet network also achieved the best performance.

Therefore, RA-ResNet is finally chosen as our feature extraction network.

3.4.2 Ablation experiment of multi steps

In order to explore and verify the effectiveness of each step of the proposed multi-step framework (MSGC-CNN), we conducted ablation experiments for each step on three datasets, Drishti-GS, RIM-ONE-R3, and ACRIMA. The experiments are as follows: (1) **Model1**: represents the basic RA-ResNet, which is directly performed on the original fundus image

Table 2 Comparison of different CNN networks and RA block in RIM-ONE-R3 dataset

Model	Accuracy (%)	Precision (%)	F1-score (%)	Kappa (%)	AUC (%)
Xception	78.12	83.45	78.91	55.20	81.36
EfficientNet	82.81	82.89	82.83	65.42	82.81
MobileNetV2	85.93	87.02	85.95	72.03	86.55
ResNet50	89.06	89.44	89.05	78.17	89.20
Xception + RA	80.65	81.24	80.60	61.41	80.83
EfficientNet + RA	84.12	88.79	84.33	68.84	87.18
MobileNetV2 + RA	87.30	87.31	87.31	74.42	87.22
ResNet50 + RA	90.47	92.15	90.49	81.12	91.43

Table 3 Comparison of different CNN networks and RA block in ACRIMA dataset

Model	Accuracy (%)	Precision (%)	F1-score (%)	Kappa (%)	AUC (%)
Xception	87.23	87.75	87.33	73.69	87.57
EfficientNet	93.61	93.64	93.62	87.02	89.16
MobileNetV2	91.49	91.76	91.46	82.91	91.33
ResNet50	95.03	95.33	95.06	89.84	95.55
Xception + RA	88.65	88.65	88.64	77.05	88.42
EfficientNet + RA	94.32	94.34	94.32	88.52	88.52
MobileNetV2 + RA	92.19	92.22	92.18	84.25	91.98
ResNet50 + RA	95.74	95.77	95.74	91.39	95.58

to diagnose the glaucoma. (2) **Model2**: represents RA-ResNet + ROI, i.e., step 1 of Fig. 1 is firstly performed on the original image to obtain the ROI image, and then RA-ResNet is performed on the ROI image to diagnose the glaucoma. (3) **Model3**: represents RA-ResNet + ROI + vessel elimination, which is based on model2, adds step 2 after step 1, eliminating the blood vessels in the ROI image, and then RA-ResNet is performed on the Seg-ROI image to diagnose the glaucoma. (4) **Model4**: represents RA-ResNet + ROI + fusion. On the basis of model 2, step 3 is added after step 1, that is, the ROI image is fused with the original fundus image. (5) **Model5**: is the proposed model in this paper, including step 1, step 2, step 3, and step 4 of Fig. 1.

We applied the above models in three datasets: Drishti-GS, RIM-ONE-R3, and ACRIMA. The data in the ACRIMA dataset are ROI images, so we only used Model2, Model3, and Model5 for experiments. Drishti-GS and RIM-ONE-R3 datasets are complete fundus images, and we used Model1, Model2, Model3, Model4, and Model5 to conduct experiments respectively.

Tables 4, 5, and 6 show the results of ablation experiments on the Drishti-GS dataset, RIM-ONE-R3 dataset, and ACRIMA dataset. From Table 4, we can draw the following conclusions: (1) The accuracy of Model2 is higher than that of Model1, which indicates that ROI extraction can improve

the accuracy of glaucoma classification. (2) The accuracy of Model3 is higher than that of Model2, which indicates that the vascular elimination of ROI can improve the accuracy of glaucoma classification. (3) The accuracy of Model4 is higher than that of Model2, which indicates that the fusion of original fundus image and ROI image can improve the accuracy of glaucoma classification. (4) Model5 is the algorithm proposed in this paper, which is the combination of ROI extraction, vessel elimination, image fusion, and RA-ResNet. By comparison, we can see that all indicators of the proposed model are higher than those of other models, which indicates that each module of the algorithm in this paper can improve the model's performance, and their combination can achieve satisfactory results.

The same conclusion can be drawn from Tables 5 and 6, and the proposed Model5 also achieves the best performance. Therefore, we can finally draw the conclusion: in the MSGC-CNN framework proposed in this paper, the steps of ROI extraction, vessel elimination, and image fusion can all improve the classification results, and the combination of them can achieve the optimal result.

Table 4 Ablations in Drishti-GS dataset

Method	Accuracy (%)	Precision (%)	F1-score (%)	Kappa (%)	AUC (%)
Model1	86.57	87.45	87.06	73.33	87.36
Model2	88.23	89.20	88.40	75.18	89.02
Model3	89.28	91.58	89.41	78.57	91.18
Model4	91.18	91.45	91.23	81.58	91.39
Model5	92.01	92.15	92.06	83.64	92.57

Table 5 Ablations in RIM-ONE-R3 dataset

Method	Accuracy (%)	Precision (%)	F1-score (%)	Kappa (%)	AUC (%)
Model1	90.47	92.15	90.49	81.12	91.43
Model2	90.62	90.85	90.65	81.10	90.87
Model3	92.06	92.09	92.05	84.07	91.97
Model4	92.18	92.25	92.19	84.28	92.27
Model5	93.75	94.48	93.75	87.54	94.12

Table 6 Ablations in ACRIMA dataset

Method	Accuracy (%)	Precision (%)	F1-score (%)	Kappa (%)	AUC (%)
Model2	95.74	95.77	95.74	91.39	95.58
Model3	96.45	96.73	96.47	92.74	97.02
Model5	97.87	97.95	97.86	95.70	97.89

3.4.3 Comparison with other methods

In this section, we compare the proposed RA-ResNet with other networks in the literatures and compare the proposed MSGC-CNN multi-step framework with other existing multi-step literatures. Our experimental parameter settings are similar to these literatures. Among them, Table 7 compares the performance of RA-ResNet and other literature networks in three datasets. Table 8 compares the proposed MSGC-CNN multi-step framework with other multi-step frameworks in three datasets.

As can be seen in Table 7, when compared to the other papers that fed fundus images directly into the network, the proposed method performs best. Among them, literature [22] employed five different ImageNet trained models for automatic glaucoma assessment using fundus images. Literature [28] proposed a new network self-ONNs for fundus images of glaucoma. Literature [29] used pre-trained GoogleNet for early diagnosis of glaucoma. Our proposed RA-ResNet50 model has been specifically

chosen and adapted to leverage the unique characteristics of fundus images. This network's depth is well-matched with the data, and we have augmented it with an additional RA functional module, designed to more effectively extract feature information from fundus photographs. Consequently, this enhanced capability enables more accurate diagnoses. Furthermore, the incorporation of transfer learning techniques strengthens the model's generalizability across various datasets. As a result, our algorithm consistently achieves superior performance on all datasets, demonstrating both high effectiveness and strong generalization capabilities. Our proposed RA-ResNet50 achieved the best results on the ACRIMA, RIM-ONE-R3, and DrishtiGS datasets with accuracies of 95.74%, 90.47%, and 86.57%, respectively. This demonstrates the proposed network designed based on fundus image characteristics can better diagnose glaucoma.

As can be seen in Table 8, when compared to the other papers of multi-step models, the proposed method performs best. The detailed analysis of the results is as follows:

Table 7 Comparison of the proposed RA-ResNet with existing CNNs in the literatures

Dataset	Methods	Accuracy (%)	Precision (%)	Kappa (%)	AUC (%)
ACRIMA	Diaz-Pinto [22]	70.21	83.33	—	76.78
	self-ONNs [28]	94.5	93.3	—	—
	RA-ResNet50	95.74	95.77	91.39	95.58
RIM-ONE-R3	Diaz-Pinto [22]	71.21	79.31	—	85.75
	self-ONNs [28]	75.3	80.5	—	—
	RA-ResNet50	90.47	92.15	81.12	91.43
Drishti-GS	Diaz-Pinto [22]	75.25	74.19	—	80.41
	Serte S [29]	75	—	—	80
	RA-ResNet50	86.57	87.45	73.33	87.36

Table 8 Comparison of the proposed MSGC-CNN with existing multi-step methods in the literatures

Dataset	Methods	Accuracy (%)	Kappa (%)	Precision (%)	AUC (%)
ACRIMA	Mañla Claro [30]	95.31	90.45	95.40	—
	Law Kumar [31]	93.65	94	95	94
	Kamesh Sonti [32]	96.70	—	95.90	96.47
	Proposed	97.87	95.70	97.95	97.89
	Cerentini [8]	86.40	—	—	—
RIM-ONE-R3	Lima [33]	79.90	—	—	86.00
	Mañla Claro [30]	91.19	82.24	91.30	—
	JAVIER [34]	90	—	—	—
	Proposed	93.75	87.54	94.48	94.12
	Orlando [35]	—	—	—	76.20
Drishti-GS	Mañla Claro [30]	86.13	64.13	86.20	—
	Pooja [36]	91.08	—	—	90.87
	Kamesh Sonti [32]	90.00	—	90.91	86.51
	Proposed	92.01	83.64	92.15	92.57

1. In the ACRIMA dataset, our accuracy reached 97.87%, which was 1.17% higher than the method proposed by Kamesh Sonti, which is the highest of the others. In terms of kappa, it reached 95.70%, which was 1.70% higher than the method proposed by Law Kumaret. In terms of precision, it reached 97.95%, 2.05% higher than the method proposed by Kamesh Sonti, which is the highest of the others.
2. In RIM-ONE-R3 dataset, our accuracy reached 93.75%, which was 2.56% higher than the method proposed by Law Kumar Singh, which is the highest of the others. In terms of kappa, it reached 87.54%, which was 5.30% higher than the method proposed by Maila Claro. In terms of precision, it reached 94.48%, 3.18% higher than the method proposed by Maila Claro, which is the highest of the others.
3. In the Drishti-GS dataset, our accuracy reached 92.01%, which was 0.93% higher than the method proposed by Law Kumar Singh, which is the highest of the others. In terms of kappa, it reached 83.64%, which was 19.51% higher than the method proposed by Maila Claro. In terms of precision, it reached 92.15%, 1.24% higher than the method proposed by Kamesh Sonti, which is the highest of the others. In terms of AUC, it reached 92.57%, which is 1.70% higher than the method proposed by Law Kumar Singh, which is the highest of the others.

According to the above statements, we can see that the proposed MSGC-CNN method achieves the best results in the three datasets, and the detailed reasons are as follows: literature [8] and [30] introduce a two-stage method that first involves cropping the image to the region of interest (ROI) and then applying a pre-trained neural network to diagnose glaucoma. Literature [31] further proposes a process that initially extracts the optic disc from fundus images using image segmentation, followed by the diagnosis of glaucoma through deep learning networks applied to the segmented optic disc images. These methods are worse than ours for not performing the process of fusion. Literature [33] applies an RGB channel weighting technique to process fundus images for glaucoma, followed by classification; however, this method shows limited improvement in the efficiency of glaucoma diagnosis. Literature [34] integrates original images with the ROI based on fusion features for diagnosis, similar to the approach in our step 4. Literature [36] presents the results of utilizing CNNs pre-trained on non-medical data for automated glaucoma detection. Literature [35] employs data augmentation and a multi-step diagnosis process involving ROI extraction followed by rotation and classification. [32] using shape and texture features to detect glaucoma. Compared with these methods, our method is better for including an

additional step of blood vessel removal and well-designed network for fundus image.

4 Conclusions

In this paper, we propose a multi-step model of MSGC-CNN, which is a new automatic glaucoma diagnosis method for the binary classification of glaucoma. In order to verify the effectiveness and superiority of the algorithm, we conducted network comparison experiments and ablation experiments on three datasets, Drishti-GS, RIM-ONE-R3, and ACRIMA and compared the results of the proposed algorithm with other state-of-the-art methods. Then, the conclusion can be drawn that MSGC-CNN can better diagnose glaucoma with more accurate results: In the network comparison experiments, we input the original fundus images into each network for glaucoma diagnosis, and the results show that the proposed RA-ResNet achieves the best classification results and demonstrate that a network specifically designed for fundus image features achieves better performance. In the ablation experiments of multi-steps, we prove that the steps of ROI extraction, vascular elimination, and image fusion can all improve the glaucoma classification results, and the combination of all modules can achieve the optimal result. In comparison with other state-of-the-art multi-step models, the proposed algorithm achieves the best performance in all three datasets, which proves the effectiveness and superiority of the proposed algorithm.

The advantages of this paper are as follows: (1) Aiming at the characteristics of small amount and high resolution of glaucoma fundus photography data, this paper proposes RA-ResNet as a feature extraction network and combines it with transfer learning technology, so as to achieve better feature extraction of glaucoma data. (2) Aiming at the problem that vascular information in ROI would interfere with glaucoma diagnosis, we use deep learning method to extract blood vessels, then eliminate the interference, and finally obtain ROI images with more effective feature information. (3) The multi-step framework MSGC-CNN in this paper combines glaucoma pathological knowledge with deep learning technology, integrates original fundus image features and ROI image features where blood vessels have been specifically removed, and finally achieves a better diagnosis of glaucoma.

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Data availability Data underlying the results presented in this paper are available in dataset 1, Ref. [21].

Data underlying the results presented in this paper are available in dataset 2, Ref. [22].

Data underlying the results presented in this paper are available in dataset 3, Ref. [23].

Data underlying the results presented in this paper are available in dataset 4, Ref. [24].

Declarations

Conflict of interest The authors declare no competing interests.

Use of AI tools declaration The authors declare they have not used Artificial Intelligence (AI) tools in the creation of this article.

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